

Denis Baldakov

AI / Machine Learning Engineer

denis1111baldakov@gmail.com | +44 7404 801567 | denisbaldakov.com | github.com/Sofa3xpert | linkedin.com/in/denis-baldakov2026

AI and Machine Learning graduate specializing in machine learning and computer vision, with internship experience across cybersecurity and enterprise AI. Focused on building and deploying ML models for automation and intelligent systems.

EDUCATION

King's College London 2023 – 2026

BSc (Hons) Artificial Intelligence — Upper Second-Class Honours (2:1)

- Relevant coursework: Machine Learning Theory, Deep Learning (CNNs, optimization), Data Science & Feature Engineering, LLMs & NLP Systems, and Cloud AI Deployment (containerization, model versioning, API-driven ML systems).

WORK EXPERIENCE

DS Corporate — AI Integration Developer (Intern) Jun 2025 – Aug 2025

Education Company

- Implemented multi-agent systems for top management to automate internal workflows.
- Conducted an audit and research for optimization of labor and introduced a new standard of work.
- Reduced manual reporting time by 40%.
- Participated in SPIEF 2025 as a company representative.

Group-IB — Anti-Fraud Analyst (Intern) Jun 2024 – Aug 2024

Global Cybersecurity Company

- Participated in an anti-fraud investigation unit in the META region.
- Analyzed real-world cyber-fraud cases: banking account takeover, authorized push payment (APP) fraud, and mule-account infrastructure.
- Researched criminal automation tooling: Selenium, PhantomJS, anti-detection browsers, AI agents.
- Improved the company's internal research base.

PROJECTS

Open-Source Contribution: HackerRank hiring-agent (7,200+ stars) — github.com/interviewstreet/hiring-agent — issue #400, PR #401

Found that HackerRank's open-source resume screener silently dropped whole sections of designed / multi-column resumes before the LLM ran; filed the issue with a minimal repro and a pull request: a one-line extraction fix plus a regression test that generates the failing layout. Status: open since Aug 2026, awaiting review, not merged. Stack: Python, PyMuPDF.

Self-Hosted LLMs for Unit-Test Generation — Dissertation — github.com/Sofa3xpert/slm-unit-test-generation
Two-agent (Programmer→Tester) pipeline benchmarking four self-hosted small language models against Claude on HumanEval and MBPP, with sandboxed execution of generated tests. Found three scoring bugs in my own harness (empty suites counted as passes, MBPP accuracy structurally zero, per-task token accounting), fixed them and re-scored every run: the headline numbers fell, and the honest comparison became the contribution. Stack: Python, LM Studio, Anthropic SDK.

Presence — local-first multi-agent job-search companion (open source) — github.com/Sofa3xpert/presence — v0.1.1, macOS app

Reads published job-board APIs on a schedule, keeps a SQLite tracker with an events timeline, briefs the user daily via their own Telegram bot, and prepares outreach in an interview-style agent session (questions → draft → per-sentence source attribution → checks). One OpenAI-compatible transport with schema-constrained JSON across Ollama, llama.cpp, LM Studio and vLLM; local engine (Nemotron 3 Nano 4B) or API keys under a hard daily budget; a human clicks every Submit. 197 tests, CI on Ubuntu / macOS / Windows. Stack: Python 3.12, Flask, pydantic, sqlite3.

YOLOv5 + MLflow Microservices — MLOps — github.com/Sofa3xpert/yolo-mlflow-microservices
Containerised object-detection platform: YOLOv5 FastAPI service, MLflow experiment tracking, MongoDB, Docker Compose, deployed to Kubernetes.

Deep Active Learning on CIFAR-10 — github.com/Sofa3xpert/active-learning-cifar10
Active-learning and semi-supervised image classification on CIFAR-10 with a ResNet backbone, comparing label-efficient selection strategies (TypiClust on supervised embeddings). Stack: PyTorch, torchvision.

Object Detection Robot — github.com/Sofa3xpert/ros-object-detection-robot
Real-time YOLO visual-recognition pipeline for a simulated robot, integrating perception with control. Stack: Python, ROS, OpenCV.

AI Mental Health Assistant — github.com/Sofa3xpert/ai-mental-health-assistant
LLM-based doctor-patient conversational assistant prototype (transformers, domain prompting). Stack: HuggingFace, BERT.

SKILLS

Machine Learning: PyTorch, TensorFlow, Scikit-learn; CNNs & Transformers, decision trees & ensembles, clustering, dimensionality reduction (PCA), GANs/VAEs, reinforcement learning, explainable AI (XAI)

Python & Data: Python, NumPy, Pandas, Plotly, OpenCV

AI / LLM: OpenAI & Anthropic SDKs, LangChain, RAG, embeddings & vector search, prompt engineering, Streamlit

Backend & MLOps: FastAPI, REST APIs, Docker, Kubernetes, MLflow

Automation & Testing: Selenium, Cypress

Databases: Cassandra, MongoDB

LANGUAGES

Russian — Native | **English** — Professional working proficiency (spoken) | **Chinese** — Basic | **German** — Basic